Mass Motion Forces

Intro to UST

Dustin Lee

Let's Start With What Is Actually Real

Mass is the only thing that truly exists. Motion and forces are just descriptions of mass. That's the starting point. Everything else in physics is either derived from those three or invented by humans to make math and measurements more manageable. Neither of those things is bad, but you need to know which is which.

Mass is stuff. It occupies ALL of space and is continuous, it has structure, and it responds with different motions and forces. Motion is what stuff does when forces act on it, it changes position over time in a direction at a rate. Forces are what one piece of mass does to another piece of mass when they interact. That's the whole foundation. Right there, in three sentences. The best way to picture this is water and air with other objects in it. All mass, just in different structures moving around in different motions that cause different forces.

Take a rock sitting on a table. You have mass, the rock, with a structure determined by how its atoms are arranged. You have forces, gravity pushing it down, the table resisting. The net force is zero, so the rock doesn't move. You push the rock. Now there's a net force in the direction you pushed. The rock moves. Done. No hidden variables. No mysterious potential wells. No need to invoke anything that doesn't physically exist. The rock, the push, and the motion are all you ever needed. Everything that follows in this paper is just an expansion of that.

Structure Is The Whole Game

Not all mass behaves the same way, and the reason is structure. An iron bar and a cloud of hydrogen gas are both mass, but they respond to forces completely differently. The iron bar transmits a force almost instantly from one end to the other. The gas cloud absorbs and disperses a force across its whole volume. Same physics. Completely different behavior. What's different? How the mass is arranged.

Structure determines how forces move through mass. Tight, rigid structure, like a steel rod or a crystal lattice, transmits force fast and direct. The atoms are packed close, their interactions are strong, and when you push one end, the push gets to the other end in a hurry. Loose, diffuse structure, like a gas, a foam, or a pile of sand, spreads force out. It dissipates. It takes longer to cross, and it arrives weaker.

This matters more than people give it credit for. When you start asking why one material behaves differently from another, the answer is always structure. Why is glass brittle? Structure. Why does rubber bounce? Structure. Why does a steel cable hold a bridge but a rope of the same diameter wouldn't? Structure. The material isn't magic. It's mass arranged a particular way, and that arrangement determines what forces it can carry and how.

This is not a new idea. Engineers have known this for centuries. We see it in magnets. What I'm saying is: stop thinking of structure as a secondary detail. It's not a detail. It's the mechanism. Structure is what turns raw mass into something that behaves a specific way. Get that straight and a lot of the confusing physics gets a lot simpler.

Forces Are Just Mass Talking To Mass

There is no action at a distance. I want to be very clear about that. When people say "gravity pulls," they're using shorthand that has caused more confusion than it should. What actually happens is this: large structured mass distorts the geometry of the mass around it. Smaller mass moving through that space follows the path that the structure makes available. The smaller mass is not being "attracted" in some mystical sense, it's moving through a space that has been physically shaped by the presence of a massive object. The mechanism is real and physical.

Contact forces are even more straightforward. When your hand pushes a wall, your hand's mass is pressing against the wall's mass. The atoms at the surface of your hand are interacting with the atoms at the surface of the wall. The wall pushes back because its structure resists compression, the atoms in the wall don't want to be pushed into each other, and the force propagates back through the wall's structure and returns to your hand. That's it. Two chunks of structured mass, exchanging force through direct physical interaction.

Forces are real. They are physical. They are not mathematical conveniences and they are not abstract relationships between abstract quantities. When a force acts, something physical is happening, one mass is pushing or pulling on another mass through some physical mechanism. If you can't tell me what that physical mechanism is, you haven't explained the force. You've just named it.

Motion Doesn't Need An Explanation, It Just Needs A Cause

Things that are moving keep moving. Not because of some law handed down from on high, but because that's just what mass does when nothing is pushing or pulling on it. Motion is the natural state of mass. Stillness is just a special case, it means all the forces on a piece of mass happen to be balanced.

When a force acts on mass, the mass changes its motion. The change is in proportion to the force and inversely proportional to the amount of mass. More mass means more force required to produce the same change. That's it. That's the whole mechanical relationship between force, mass, and motion. No mysteries buried in there.

You do not need energy to explain any of this. Mass, force, and distance are sufficient. You have a mass. A force acts on it for some duration. The mass ends up moving differently than before. That's the complete story. The concept of "energy" adds nothing to your understanding of the mechanism. It gives you a useful number you can track across a complex system, but it doesn't tell you why the mass moved or what physically caused the change. Mass caused it. Force caused it. Time was the duration. Simple.

People sometimes argue that without energy, you can't account for the capacity to do work a moving object has. But look at what that phrase actually says. A moving object has mass and it has motion. When that object collides with something else, its mass and its motion, transmitted through force on contact, change the motion of the other object. That's what doing work looks like, mechanically. The phrase "capacity to do work" is just a label for "mass moving at a certain rate." The label is fine. Treating the label as a cause is where things go sideways.

Energy Is A Receipt, Not A Cause

Nobody has ever held a joule. You cannot point to energy in a room. You cannot put energy in a box and weigh it. What you can do is point to mass moving at some rate, or mass compressed against other mass, or mass whose internal parts are vibrating faster than before. Those are real things. Those are what energy measures. Energy is the receipt you get after the physical transaction has already happened.

When we say, "energy is transferred from object A to object B," here is what actually occurred: mass moved, forces acted between the two objects, and at the end of the interaction, different mass is moving differently than before. The energy numbers shifted because energy is defined in terms of mass and motion, so when mass and motion change, the energy number changes. The energy number being conserved is not a miraculous fact about the universe, it's a consequence of how we defined the term. We built conservation in. Of course it conserves. That's circular.

None of this means the equations are wrong. They're not wrong. $F = ma$ works. Kinetic energy formulas work. Conservation of energy works as a calculation tool. The mistake, and it's a big one, is treating the variable in the equation as though it's a real substance that causes other things. Energy does not cause motion. Forces cause motion. Energy is the scorecard.

Think of it this way. When you look at your bank account balance, that number doesn't cause you to buy groceries. You buy groceries because you need food and you go to a store and you hand over money. The bank balance is a record of transactions. It tracks what happened. It doesn't make anything happen. Energy is the bank balance of the physical universe. Useful for bookkeeping. Not the thing doing the work.

EM Is Just Matter Being Pushed Through Structured Mass

A wire is a tube of structured mass. The atoms in a copper wire are arranged in a lattice that allows certain particles, specifically electrons, to be shoved along the lattice from one end to the other when you create a pressure difference between the two ends. Voltage is that pressure difference. It's not a mystical potential. It's a push. Higher pressure on one side, lower on the other, particles move toward the low-pressure end. Every plumber on earth understands this concept.

When those electrons move through the wire, they push on the mass around the wire. That push propagates outward from the wire in all directions. It looks like a "field" because you have a continuous motion along the wire producing a continuous push in the surrounding space. But there is no field in the sense of an invisible non-material entity hovering in space. There is mass, the electrons, moving, and their motion physically influencing nearby mass through mechanical interaction. It's a push-pull situation, continuously maintained as long as the electrons keep moving. The vacuum is not empty. It is mass and it moves because of the motion of the electrons.

A magnet is structured mass whose internal arrangement is locked into a consistent orientation. The atoms, or more precisely, the subatomic structure of the iron, are all pointing the same direction. That consistent directional arrangement produces a consistent directional push on any nearby mass that can be influenced by that kind of interaction. It's not mysterious. The structure of the magnet is doing mechanical work on nearby mass. That's all magnetism is: structured mass pushing on other mass in a directional way because of how it's internally arranged. Same idea as wind or wave.

Put a wire near a magnet and run current through the wire. The moving electrons in the wire are now passing through a region where the surrounding mass, the magnet, is already exerting a directional push. The magnet's structured mass deflects the path of the moving electrons. The electrons push back on the wire. The wire moves. This is a motor. It is a physical push from one piece of mass on another piece of mass, mediated by the internal structure of both. There is nothing in that description that requires you to believe in a magical invisible entity called a field.

And light? Light is mass moving fast in a structured wave pattern. Whether you call it a wave or a particle, something physical is traveling through mass or space. That something has the capacity to push on other mass when it arrives, because it has mass and motion, and mass in motion can exert a force on contact. The dual nature of light isn't a philosophical paradox. It's just what you get when you're looking at a very small, very fast piece of structured moving mass from different angles that travels at different speed depending on friction of the surrounding mass. This friction is heat.

Why This Framework Is Simple

The Mass-Motion-Forces framework has one primary advantage over the standard way physics gets taught and talked about: it never asks you to believe in anything you can't physically point to or picture in your head.

No fields. No potentials. No wavefunctions. No probability density clouds floating in abstract Hilbert space. No mysterious dark energy filling the vacuum. Just mass, which exists and has structure, moving at some rate in some direction, exchanging forces with other mass through physical contact or proximity. That's the whole ontology. You can build everything else from it.

It is fully mechanistic. At any moment in any physical process, you can ask: what is physically happening right now? And you get a real answer. Mass is here. It is moving this direction at this rate. This force is acting on it from that direction because that piece of mass over there has this structure and is interacting with it. No hand-waving. No "well, you have to accept that some things just work this way at the quantum level." No. Tell me what's moving. Tell me what's pushing on what. If you can't do that, you haven't explained it yet.

The framework scales. Whether you're talking about galaxies colliding or subatomic particles scattering, the same three things are present: mass with structure, moving at some rate, exchanging forces with other mass. The scale changes. The complexity of the structure changes. The specifics of the interaction change. The framework doesn't change.

And importantly, the math still works. This framework does not require throwing out a single equation. Newton's laws still apply. Maxwell's equations still apply. The equations are good descriptions of what mass, motion, and forces do. The only thing this framework changes is what you think the variables in those equations represent. Stop treating them as causes. Start treating them as descriptions. The numbers stay the same. Your understanding of what they mean gets a lot cleaner.

The Takeaway

The universe is a machine. It is made of stuff. The stuff moves. Moving stuff pushes on other stuff. That is the whole story. Mass built the universe. Forces shaped it. Motion is the result of those forces acting on that mass. Energy did not build anything. Energy is what you calculate after the building is done. If you understand this you can engineer mass into structures that direct the flow of the matter around it. The possibilities are almost unlimited.

The reason physics feels mysterious to most people is not because the universe is mysterious, it's because the language used lacks the mechanism that people can connect the words to. Fields, potential energy, wave-particle duality, quantum uncertainty are descriptions of something we can't see. What's happening is always mass, motion, and forces. Every single time.

The next time someone explains something by saying "the energy caused this" or "the field did that," ask them two questions: What mass moved? And what force acted? Those are the questions that get you to a real answer. The energy and the field are just ways of naming the situation after it's already been caused by mass and forces. The rest is details, but the devil is in the details.

Math and Measurements

Change in entropy per change in description

$$\frac{\Delta S}{\Delta D} = k$$

Force or energy equals mass times squared motion or speed

$$\frac{E}{mc^2} = 1$$

Ratio of mass-force scale to configuration scale

$$\frac{5.94728569105370 \times 10^{-27}}{5.15484850640341 \times 10^{96}} = 1.15372657094887 \times 10^{-123}$$

Logarithmic scale of that ratio

$$\log_{10}\left(\frac{5.94728569105370 \times 10^{-27}}{5.15484850640341 \times 10^{96}}\right) = -122.93789710521712$$

Entropy change from change in configuration freedom

$$\Delta S = k_B \Delta(\ln \Omega)$$

Entropy change per unit description change

$$\frac{\Delta S}{\Delta D} = \frac{k_B \Delta(\ln \Omega)}{\Delta D} = 1.380649 \times 10^{-23}\ \mathrm{J} \cdot \mathrm{K}^{-1}$$

Entropy increment per nat

$$\Delta D = 1 \ \Rightarrow \ \frac{\Delta S}{\Delta D} = k_B$$

Entropy increment per bit

$$\Delta D = \ln 2 \;\; \Rightarrow \;\; \frac{\Delta S}{\Delta D} = k_B \ln 2$$

Motion-spread per mass-arrangement change

$$\frac{\Delta(\text{motion-spread})}{\Delta(\text{mass-arrangement})} = k_{\text{motion}}$$

Motion-spread produced by mass-arrangement change

$$\Delta(\text{motion-spread}) = k_{\text{motion}} \cdot \Delta(\text{mass-arrangement})$$

Motion content stored in mass

$$M_{\text{motion}} = mc^2$$

Fraction of motion relative to force capacity

$$R = \frac{\text{mass-motion}}{\text{force-capacity}}$$

Logarithmic force-capacity scale

$$\text{force-scale} = \log_{10}\left(\frac{\text{mass-motion}}{\text{force-capacity}}\right)$$

Motion-spread from configuration freedom

$$\Delta(\text{motion-spread}) = k_{\text{motion}} \cdot \Delta(\text{mass-configuration freedom})$$

Motion-spread per nat

$$\frac{\Delta(\text{motion-spread})}{\Delta(\text{mass-arrangement})} = k_{\text{motion}}$$

Motion-spread per bit

$$\frac{\Delta(\text{motion-spread})}{\Delta(\text{mass-arrangement})} = k_{\text{motion}} \ln 2$$

Unified set

$$\Delta(\text{motion-spread}) = k_{\text{motion}} \cdot \Delta(\text{mass-arrangement})$$

$$M_{\text{motion}} = mc^2$$